Graph-Structured Trajectory Extraction from Travelogues

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Abstract

Previous studies on *sequence*-based extraction of human movement trajectories have an issue of inadequate trajectory representation. Specifically, a pair of locations may not be lined up in a sequence especially when one location includes the other geographically. In this study, we propose a *graph* representation that retains information on the geographic hierarchy as well as the temporal order of visited locations, and have constructed a benchmark dataset for graph-structured trajectory extraction. The experiments with our baselines have demonstrated that it is possible to accurately predict visited locations and the order among them, but it remains a challenge to predict the hierarchical relations.

1 Introduction

Travelogues are a significant source for analyzing human traveling behavior in various fields, including tourism informatics (Akehurst, 2009; Hao et al., 2010) and human geography (Cooper and Gregory, 2011; Caselli and Sprugnoli, 2021), because of their rich geographic and thematic content, which gives people, for example, a simulated experience of trip (Haris and Gan, 2021). In particular, human traveling trajectories play a central role in characterizing each travelogue, and thus, automatic trajectory extraction from travelogues is highly desired for tourism services, such as travel planning and recommendation (Pang et al., 2011).

Some studies have addressed automatic trajectory extraction from text (Ishino et al., 2012; Wagner et al., 2023; Kori et al., 2006). However, these studies have two problematic issues: (i) inadequate trajectory representation and (ii) the scarcity of benchmark datasets.

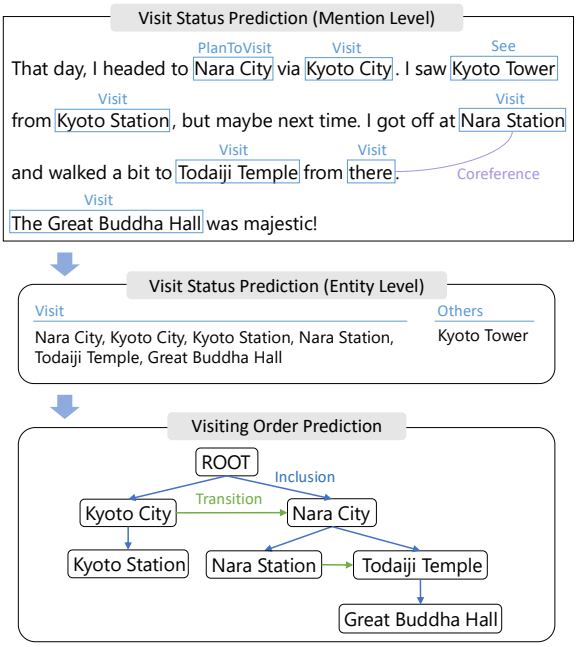


Figure 1: Illustration of our proposed tasks: visit status prediction (VSP) and visiting order prediction (VOP). The goal of VSP is to assign visit status labels to mentions (top) and to entities (middle). The goal of VOP is to build a visiting order graph by identifying inclusion and transition relations between entity pairs (bottom).

First, the previous studies treated each trajectory as a *sequence* of visited locations (Ishino et al., 2012; Wagner et al., 2023; Kori et al., 2006), but a sequence is inadequate as a representation of trajectories. This is because a pair of locations may not be lined up in a single sequence especially when one location includes the other geographically, for example, "Kyoto City" and "Kyoto Station."

Second, the previous studies constructed and used their in-house datasets for evaluating their systems, and no public text datasets have been released with trajectory information annotation. However, shared benchmark datasets are necessary for facilitating fair comparisons with other

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studies and accelerating the accumulation of research findings (Ohsuga and Oyama, 2021).

For the first issue, we propose a *visiting order graph* illustrated at the bottom of Figure 1. This graph has location or geo-entity nodes and relation edges between nodes, and can represent not only temporal *transition relations* but also geographic *inclusion relations* between visited locations. For facilitating automatic graph construction for each travelogue, we introduce subtasks of trajectory extraction: Visit Status Prediction (VSP) and Visiting Order Prediction (VOP), as shown in Figure 1. VSP assigns *visit status labels* to mentions and entities, while VOP identifies inclusion and transition relations between nodes of the "visited" entities.

For the second issue, we have constructed a dataset for training and evaluating trajectory extraction systems: Arukikata Travelogue Dataset with Visit Status and Visiting Order Annotation (ATD-VSO).[1] Our dataset comprises 100 travelogue documents annotated with the corresponding visiting order graphs, including 3,354 geo-entities (nodes) and 3,369 relations (edges) in total.

Using this dataset, we have trained and evaluated baseline systems. Notable findings through the experiments are (i) that the systems can achieve relatively high accuracy for predicting visit status labels and the transition relations, and (ii) that the systems failed to accurately predict inclusion relations. The latter implies an important future investigation, i.e., how to inject the knowledge of geographic hierarchical structure into the systems.

Contributions For the purpose of building a foundation for future studies, we have made two main contributions: (i) the proposal of a visiting order graph and (ii) the construction of a benchmark dataset for the trajectory extraction.[2] We will release our code and dataset for research purposes. We expect that our dataset will foster continued growth in the trajectory extraction research.

2 Preliminaries for Data Construction

Our dataset, ATD-VSO, has been constructed on the basis of ATD-MCL (Higashiyama et al., 2023),[3] which is a Japanese travelogue dataset annotated with geo-entity information, including mentions and their coreference relations, using a subset of the original travelogues, the Arukikata Travelogue Dataset (Arukikata. Co., Ltd., 2022; Ouchi et al., 2023).[4] Annotated mentions in ATD-MCL include proper nouns (e.g., "Nara station"), general noun phrases (e.g., "the station"), and deictic expressions (e.g., "there") that refer to various types of locations, such as geographic regions, facilities, and landmarks. Moreover, a set of mentions that refer to the same location constitutes a coreference cluster or geo-entity.[5] Given such annotated travelogues, we focus on annotating the visit status and visiting order of candidate geo-entities.

3 Visit Status Prediction

In this section, we describe the task definition, annotation data construction, and our baseline system for Visit Status Prediction (VSP), which is the first step of trajectory extraction with given mentions and entities. The goal of VSP is to identify locations that the traveler[6] visited by assigning a visit status for each location mentioned in the travelogue. For example, it is possible to judge that the traveler actually visited the station from the description of the real experience: "Arrived at Kintetsu Nara Station!" In contrast, the following factual statement does not indicate whether the traveler visited these stations or not: "JR Nara Station is a little far from Kintetsu Nara Station."

3.1 Annotation Data Construction

We defined two types of visit status labels for entities and six types of visit status labels for mentions in Table 1.[7] The mention labels serve to distinguish detailed status for the mentioned locations based on the context, which typically corresponds to the sentence where the mention occurs. The entity labels serve to determine whether the traveler eventually visited the location, considering the context of the entire document.

As annotation work, native Japanese annotators at a data annotation company assigned visit status

[1] We will release our dataset at ANNONYMIZED_URL.

[2] Our contributions are in the data resource direction, not the technical one such as algorithm and model sophistication. On top of the resource, we will make technical contributions in the future.

[3] http://github.com/naist-nlp/atd-mcl

[4] https://www.nii.ac.jp/dsc/idr/arukikata/

[5] In this paper, we use the term "geo-entity" or "entity," instead of "coreference cluster."

[6] We assume that the traveler in a travelogue usually corresponds to the writer of the travelogue.

[7] Actual examples are listed in Table 9 in Appendix A.

	Label	Definition
1	`Visit`	A visit to the location is stated or implied.
2	`Other`	Not 1.
1	`Visit`	The same as the label `Visit` for entities.
2	`PlanToVisit`	It mentions a plan to visit the location during this trip (described in the travelogue).
3	`See`	Not any of 1–2, and that the traveler saw the location can be identified.
4	`Visit-Past`	Not any of 1–3, and it mentions having visited the location before this trip.
5	`Visit-Future`	Not any of 1–3, and it mentions the intention to visit the location after this trip.
6	`UnkOrNotVisit`	The visit to the locations cannot be identified from the descriptions, or the non-visit can be identified.

Table 1: Visit status labels for entities (top) and mentions (bottom).

Set	#Doc	#Sent	#Men	#Ent	#Inc+Tra
Train	70	4,254	3,782	2,339	2,343
Dev	10	601	505	316	329
Test	20	1,469	1,102	699	697
Total	100	6,324	5,389	3,354	3,369

Table 2: Basic statistics of the ATD-VSO: the numbers of documents ("#Doc"), sentences ("#Sent"), mentions ("#Men"), entities ("#Ent"), and inclusion and transition relations ("#Inc+Tra").

labels to each mention and entity in ATD-MCL travelogues using the brat annotation tool (Stenetorp et al., 2012),[8] according to the label definitions and annotation guideline.

Inter-Annotator Agreement We requested two annotators to independently annotate five documents. We then measured the inter-annotator agreement (IAA) using F1-score and Cohen's Kappa κ. The obtained scores suggest the high agreement: F1 score of 0.80 and κ of 0.68 for 180 mentions, and F1-score of 0.89 and κ of 0.81 for 124 entities.

Data Statistics 95 documents were allocated among multiple annotators, and one annotator annotated each document. The total became 100 documents, including the aforementioned five documents, with total 3,354 entities. For the constructed dataset, the basic statistics and detailed statistics are shown in Table 2 and Table 10 in Appendix B, respectively.

3.2 Task Definition

Entity-level and mention-level VSP are defined as follows: Given a set of entities $\mathcal{E}$ in an input document, entity-level VSP requires a system to assign an appropriate visit status label $y \in \mathcal{L}_e$ for each entity $e_q \in \mathcal{E}$. Similarly, given an entity (or coreference cluster) $e_q = \{m_1^{(q)}, \ldots, m_{|e_q|}^{(q)}\}$, which consists of one or more mentions, mention-level VSP requires a system to assign an appropriate visit status label $y \in \mathcal{L}_m$ for each mention $m_i^{(q)} \in e_q$.

3.3 Baseline System

As our baseline system, we employ a two-step method that first predicts mention labels and then predicts entity labels based on the mention labels. Specifically, we calculate the label probability distribution $P(y|m_i^{(q)})$ for each mention $m_i^{(q)} \in e_q$, and select the most probable label $\hat{y}_i^{(q)}$:

$$\hat{y}_i^{(q)} = \underset{y \in \mathcal{L}_m}{\arg\max}\ P(y|m_i^{(q)}).$$

Then, we select a label for each entity e_q according to the following mention label aggregation (MLA) rules.

1. If `Visit` or `PlanToVisit` has been assigned to at least one mention in e_q, then `Visit` is assigned to e_q.
2. Otherwise, `Other` is assigned to e_q.

Model Implementation As the implementation of a model for mention label prediction, we used `LukeForEntityClassification` in Hugging Face Transformers[9] with the inputs of the sentence containing the mention of interest and the position (character offsets) of the mention.

4 Visiting Order Prediction

In this section, we first define a visiting order graph, which is a key concept for our structured trajectory extraction. Then, we describe the task definition, annotation data construction, and

[8] https://github.com/nlplab/brat

[9] https://huggingface.co/docs/transformers/index

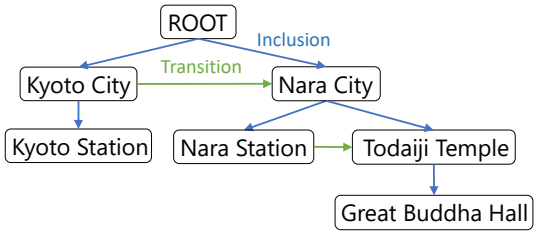


Figure 2: Example of a visiting order graph, the same example at the bottom of Figure 1.

our baseline system for Visiting Order Prediction (VOP), which involves prediction of geographic and temporal relations between visited locations.

4.1 Visiting Order Graph

We introduce a visiting order graph to represent non-linear relations of visited locations. Figure 2 shows an example. Nodes correspond to entities, and edges correspond to relations between entities. In Figure 2, blue vertical/diagonal directed edges are inclusion relations indicating the starting node geographically includes the ending node, and green horizontal directed edges are transition relations indicating the traveler visited the starting node entity and then directly visited the ending node entity, without visiting any other entities in between.

Inclusion Relation Consider the example document in Figure 1, which describes that the traveler visited both "Nara City" and "Todaiji Temple." Based on the geographic fact that the region of "Nara City" includes the region of "Todaiji Temple," a reasonable interpretation is that the traveler visited the temple and thereby also visited the city simultaneously. We introduce inclusion relation $\langle e_1, e_2 \rangle$, where an entity e_1 geographically includes another e_2. We pick up the following two instances from the graph in Figure 2:

$$p_1 = \langle \text{Nara City}, \text{Todaiji Temple} \rangle,$$
$$p_2 = \langle \text{Todaiji Temple}, \text{Great Buddha Hall} \rangle.$$

Here, p_1 indicates that "Nara City" geographically includes "Todaiji Temple", and p_2 indicates that "Todaiji Temple" includes "Great Buddha Hall." From these two relation instances, we can deduce a hierarchical relation that "Nara City" is a grandparent of "Great Buddha Hall."

Transition Relation Given a set of entities for a document and inclusion relations among them, we assign transition relation to each pair of preceding and subsequent visited entities. Notably, we restrict an entity pair with transition relation to two entities with the same parent entity. In Figure 2, whereas "Nara Station" and "Todaiji Temple" have the same parent node, "Kyoto Station" and "Nara Station" do not. Therefore, the transition relation can be assigned to $\langle$Nara Station, Todaiji Temple$\rangle$, but cannot be assigned to $\langle$Kyoto Station, Nara Station$\rangle$. This restriction allows us to determine the order of visits for any entity pairs by traversing transition and inclusion relations, even if entity pairs are not directly related to each other. For example, "Kyoto Station" does not have transition relation to "Nara City," but we can infer that "Kyoto Station" was visited before "Nara City" because the parent "Kyoto City" has transition relation to "Nara City."

Other Criteria Visiting order graphs with the two relation types above can represent various trajectories, but not all. We further introduce the following criteria for exceptional cases.

- Multiple Visits: There can be entities revisited after passing through other entities. Such an entity should be split into sub-entities that include the corresponding mentions for each visit, and sub-entities are regarded as nodes in a graph instead of the original entity.
- `UnknownTime`: There can be entities to which the timing of the visit is ambiguous. Such an entity should be assigned the `UnknownTime` label and is excluded from nodes in a graph.
- `Overlap`: There can be two entities geographically overlapping each other, but one does not include the other, e.g., "Honshu" and "Tokyo Prefecture."[10] Such two entities should be assigned the `Overlap` relation, and either entity can be selected as a representative node to be assigned inclusion and transition relations between it and other entities.

4.2 Annotation Data Construction

After the visit status annotation step, only entities with the `Visit` label were retained as the nodes of a visiting order graph and used as input for a successive relation annotation step. In this step, annotators assigned relations between the entities

[10] Honshu is the largest island that constitutes the mainland Japan. Tokyo Prefecture is a geopolitical region that consists of a part of Honshu and 11 islands.

using an online whiteboard service, Miro,[11] which can be served as an annotation tool with GUI.

Inter-Annotator Agreement We requested two annotators to independently annotate the same five documents as those used for visit status annotation. We then measured the IAA using F1-score. The obtained F1 scores suggest the moderate or high agreement: 0.94 for inclusion, 0.74 for transition, and 0.85 for both.

Data Statistics The 95 documents already annotated with visit status were allocated among five annotators, and one annotator annotated each document. The total became 100 documents, including the aforementioned five documents, with 1,863 inclusion relation and 1,506 transition relation instances in total. The detailed statistics are shown in Table 11 in Appendix B.

4.3 Task Definition

The task of VOP involves two subtasks: Inclusion Relation Prediction (IRP) and Transition Relation Prediction (TRP).

Inclusion Relation Prediction Given a set of entities $\mathcal{E}$ in a document, IRP requires a system to determine the parent entity for each entity $e_q \in \mathcal{E}$ from the set of candidate entities $\mathcal{P}^{(q)}_{\text{cand}} = \mathcal{E} \setminus \{e_q\} \cup \{\texttt{ROOT}\}$. If $e \in \mathcal{P}^{(q)}_{\text{cand}}$ is predicted as the parent entity for e_q, it means that e includes e_q. The pseudo parent node ROOT should be predicted when the entity of interest has no parent entities.

Transition Relation Prediction Given a set of entities $\mathcal{E}$ in a document, TRP requires a system to determine the entity subsequently visited for each entity $e_q \in \mathcal{E}$ from the candidate set $\mathcal{S}^{(q)}_{\text{cand}}$ with the same parent as that of e_q: $\mathcal{S}^{(q)}_{\text{cand}} = \{e_k \in \mathcal{E} \,|\, e_k \neq e_q \,|\, \text{Par}(e_k) = \text{Par}(e_q)\} \cup \{\texttt{EOS}\}$. Here, $\text{Par}(e)$ represents the parent entity of e, and the pseudo subsequent node EOS (End of Sequence) represents that the entity of interest has no subsequent entities.

4.4 Baseline System

The baseline systems adopt similar methods for the two subtasks. Specifically, for IRP and TRP, we select the most probable entity as the parent entity $\hat{e_p}$ and the subsequent entity $\hat{e_s}$ from the corresponding candidate set based on score functions $\text{score}_{\text{par}}$ and $\text{score}_{\text{sub}}$, respectively:

$$\hat{e_p} = \underset{e' \in \mathcal{P}^{(q)}_{\text{cand}}}{\arg\max} \ \text{score}_{\text{par}}(e_q, e'), \quad (1)$$

$$\hat{e_s} = \underset{e' \in \mathcal{S}^{(q)}_{\text{cand}}}{\arg\max} \ \text{score}_{\text{sub}}(e_q, e'). \quad (2)$$

Model Implementation To implement models with the score functions for both IRP and TRP, we used `LukeForEntityPairClassification` (Yamada et al., 2020) in Hugging Face Transformers, which receives the input text and the positions (character offsets) of two mentions. The input text is constructed by concatenating the two sentences containing representative mentions for the target entity e_q and a candidate entity e' and all the sentences that occur between them, in the order of their occurrence.[12]

Representative Mention Selection For IRP, proper noun mentions are prioritized over others and selected as representative ones. For TRP, we define the priority for visit status labels, i.e., `Visit` > `See` > other labels, and select the mentions with higher priority as representative ones.[13]

Sequence Sorting Decoding For TRP, all sibling nodes sharing the same parent node should be arranged in a single sequence. However, naïve decoding based on predicted pairwise scores (Equation 2) does not necessarily satisfy this requirement. To address this issue, we introduce sequence sorting decoding based on a greedy search strategy as follows.

1. Let $\mathcal{P}$ a set of all possible pairs whose nodes have the same parent.
2. Select a pair $\langle e_a, e_b \rangle$ with the highest score from $\mathcal{P}$.
3. Exclude all pairs conflicting with $\langle e_a, e_b \rangle$ from $\mathcal{P}$, namely, $\langle e_b, e_a \rangle$, $\langle *, e_b \rangle$ and $\langle e_a, * \rangle$. Here, "$*$" indicates an arbitrary node.
4. If transition relations among all the nodes have been determined, terminate the decoding. Otherwise, return to Step 2.

[11] https://miro.com/

[12] For example, when 'Nara City' and 'Todaiji Temple' in Japanese translation of the first three sentences in Figure 1 are entities of interest, the input text is represented as "<s><ent><ent></s><s>...</s><s><ent2><ent2></s>".

[13] For both subtasks, preceding mentions are prioritized if there are multiple mentions with the same priority.

5 Experiments

We evaluated the performance of our baseline systems and basic rule-based systems for the visit status prediction (VSP) task (§3.3) and the visiting order prediction subtasks: inclusion relation prediction (IRP) and transition relation prediction (TRP) (§4.4).

5.1 Experimental Settings

Data Split As shown in Table 2, we split the 100 documents in ATD-VSO into training, development, and test sets at a ratio of 7:1:2.

Task Settings We adopted the settings where gold labels of preceding tasks were given to systems, and evaluated the systems for each task independently. That is, systems take gold entities for VSP, the gold entities with gold visit status labels for IRP, and gold visited (sub-)entities[14] and gold inclusion relations for TRP, respectively, as inputs. The main reason for the gold label settings for all the tasks is that we focus on investigating the system performance for each task independently. We leave more advanced investigations, including evaluation of end-to-end models, for future work.

Evaluation Metrics For VSP, we measured the accuracy of predicted labels for input entities as well as macro-averaged F1 score over all labels. For IRP, we measured the F1 score for extracting inclusion entity pairs from input entities. For TRP, we measured the F1 score for extracting transition entity pairs, excluding pairs where the subsequent entity is `EOS`.

Model Training We constructed our baseline system by fine-tuning a pretrained model with the training set for each task. Specifically, we used a pretrained multilingual LUKE (Ri et al., 2022) model[15] for VSP and the same pretrained Japanese LUKE (Yamada et al., 2020) model[16] for IRP and TRP. Unless otherwise specified, we report the mean accuracy or F1 score on the test set of five runs with different random seed values for the baseline system for each task. More details on model training settings are shown in Appendix C.

[14]For TRP, we adopt a setting where the number of visits for each entity is given.

[15]https://huggingface.co/studio-ousia/mluke-large-lite

[16]https://huggingface.co/studio-ousia/luke-japanese-base

Level	Method	Acc.	Macro F1
Mention	Majority Label	0.679	0.135
	LUKE	**0.789**	**0.468**
Entity	Majority Label	0.823	0.451
	LUKE + MLA	**0.862**	**0.740**

Table 3: System performance for visit status prediction.

Label	Mention			Entity		
	P	R	F1	P	R	F1
`Visit`	.835	.913	.872	.896	.942	.918
`Plan`	.706	.688	.696	–	–	–
`See`	.655	.661	.657	–	–	–
`Past`	0	0	0	–	–	–
`Future`	0	0	0	–	–	–
`UN/O`	.611	.403	.482	.650	.495	.561

Table 4: Precision (P), Recall (R), and F1 scores of LUKE (mention-level) and LUKE+MLA (entity-level) for each label of visit status prediction.

5.2 Results for Visit Status Prediction

Systems We evaluated a rule-based system (Majority Label) and baseline systems (LUKE and LUKE+MLA). The Majority Label rule always outputs the most frequent label, `Visit`, for both mention and entity levels. LUKE indicates the baseline system, and MLA indicates the mention label aggregation rule described in §3.3.

Results Table 3 shows the performance for mention-level and entity-level VSP. Majority Label achieved good accuracy (0.679 for mentions and 0.823 for entities), indicating the imbalance in label distribution with a majority of `Visit` instances. This aligns with the intuition that most of the locations mentioned in travelogues are visited ones. The baselines, i.e., LUKE and LUKE+MLA, achieved better performance, owing to the capability of modeling both mention and context information, as discussed in detail in Appendix D.1 (Table 13).

Label-Wise Performance Table 4 shows the performance of the LUKE baselines for each label. First, the baselines achieved high performance (F1 of 0.872–0.918) for `Visit` for both levels. Second, the baselines resulted in limited performance (F1 of 0.482–0.561) for `UnkOrNotVisit/Other`. The confusion matrices (Figure 4 and 5 in Appendix D.1) shows the most frequent errors were misclassification of `UnkOrNotVisit/Other`

Depth	#Ent	F1		
		Random	Flat	LUKE
All	468	0.043	0.244	**0.355**
1	114	0.057	**1**	0.058
≥2	354	0.038	0	**0.425**

Table 5: System performance (F1 score) for inclusion relation prediction. Depth indicates the distance to ROOT of entity nodes based on the gold inclusion hierarchy.

Method	All	Fwd.	Rev.
Random	0.190	0.247	0.061
OccOrder (early mention)	0.730	0.773	0
OccOrder (visit status)	**0.758**	0.794	**0.386**
LUKE (naïve score)	0.680	0.737	0.298
LUKE (sequence sorting)	0.748	**0.796**	0.366

Table 6: System performance (F1 score) for transition relation prediction. "All" indicates the performance for all entities. "Fwd." and "Rev." indicate the performance for entities whose gold subsequent entities occurred after or before the entities of interest in documents, respectively, regarding their earliest mentions.

as Visit.

Directions for Improvement The LUKE baselines take limited context as input. LUKE predicts a label for each mention based on the context within the sentence where the mention occurs, and the MLA rule predicts a label for the entity by merely aggregating the mention-level predictions. A possible direction to improve entity-level performance is to construct an end-to-end model that integrates information on all mentions for an entity with the wider context of the document.

5.3 Results for Inclusion Relation Prediction

Systems We evaluated two rule-based systems (Random and Flat) and a baseline system (LUKE). Random is a rule-based system that randomly selects the parent entity from the candidate set for each entity. Flat is the other rule-based system that always selects ROOT as the parent entity for an arbitrary entity. LUKE indicates the baseline system described in §4.4.

Results Table 5 shows the performance (F1 score) of the evaluated systems for IRP. Flat exhibited the better performance than Random (F1 of 0.244 vs 0.043), suggesting that predicting ROOT can be a reasonable strategy when systems do not have knowledge for specific entities. LUKE achieved the best performance (F1 of 0.355) and, in particular, outperformed Random by a large margin for the entities whose gold parents are entities other than ROOT (depth≥2). By contrast, LUKE showed poor performance similar to Random for the entities whose gold parents are ROOT (depth=1). Detailed performance for each depth is shown in Table 14 in Appendix D.2.

Directions for Improvement The current LUKE baseline has two limitations. First, the absolute overall performance (F1 of 0.355) has substantial room for improvement. Potential reasons for the limited performance are that (1) the pretrained LUKE model for general entity analysis tasks did not learn geographic relations among specific geo-entities, and (2) it was difficult to obtain generalized knowledge on geographic relations between entities from fine-tuning only with text-based features. Possible solutions include pretraining with geospatial information like GeoLM (Li et al., 2023), and fine-tuning a model with geocoding-based features, such as predicted coordinates and shapes of entities. Second, the performance is quite low for entities whose parent is ROOT. This is because the baseline predicts ROOT as the parent for an entity only when it predicts all candidate entities as non-parent. This may be improved by a method that can directly predict ROOT by assigning a vector representation to ROOT based on a dummy sentence.

5.4 Results for Transition Relation Prediction

Systems We evaluated three rule-based systems (Random and two variants of OccOrder) and two variants of the LUKE baseline system. The Random rule randomly lines up candidate entities for each set of entities with the same parent entity. OccOrder arranges candidate entities in the order of occurrence of each representative mention in their document; whereas the "early mention" strategy (OccOrder-EM) uses the earliest occurrence mention as the representative mention, the "visit status" strategy (OccOrder-VS) prioritizes mentions based on visit status label similarly to LUKE (§4.4). The LUKE variants correspond to the baseline system with naïve score-based decoding and sequence sorting decoding (§4.4).

Results Table 6 shows the performance of the evaluated systems for TRP. For all pairs, the Oc-

SentID	Original Japanese Text	English Translation
025-02	G:UN P:Vis G:UN P:Vis	This time, I will skip Matsue G:UN P:Vis and Yonago G:UN P:Vis .
027-01	G:UN P:Vis	Matsue Shinjiko Onsen Station G:UN P:Vis is the final station.
046	G:Plan P:Plan 1 G:Vis P:Vis	I planned to stay one night in Nagoya G:Plan P:Plan , so I left Ise G:Vis P:Vis even though it was still early.
024	G:Vis P:Vis	Here is Daiouji Temple G:Vis P:Vis with its mausoleum.
025	G:Vis P:Vis	I took a taxi because it was far from the station G:Vis P:Vis .

Table 7: Actual sentences in documents: ID 23251 (top), 20816 (middle), and 20851 (bottom). The superscript "G" to mentions indicates gold visit status labels and the subscript "P" indicates LUKE's prediction (`Vis` for `Visit`, `Plan` for `PlanToVisit`, and `UN` for `UnkOrNotVisit`).

cOrder variants achieved relatively high performance (F1 of 0.730–758). This matches the intuition that the order of locations being described in text corresponds with the order of locations being visited to some extent. LUKE with the decoding based on its raw pairwise scores underperformed the OccOrder variants. However, LUKE with the sequence sorting decoding improved approximately 0.07 F1 points and achieved performance on par with that of OccOrder-VS. This result shows the effectiveness of the sequence sorting decoding. We investigated another aspect related to the prediction difficulty: the size of the candidate entity set might have an impact on the drop of the Random performance as observed in Table 15 in Appendix D.3. However, no clear tendencies were observed for OccOrder and LUKE.

Directions for Improvement The current LUKE baseline has two limitations. First, the vector representation of an entity is constructed from a single mention selected by the heuristic rule (§4.4), which limits the context of the entity. This would be improved by extending the context to include all mentions for two entities of interest, although it is necessary to explore an effective method that can grasp complicated relations among many mentions. Second, the current baseline uniformly treats entity pairs without transition relation as negative instances. However, entity pairs with indirect transition relation, where one is visited before the other via one or more entities, can be exploited as positive instances for an auxiliary task, similarly to relative event time prediction (Wen and Ji, 2021).

6 Error Analysis

We investigated tendencies of prediction errors of the current baseline systems: LUKE+MLA for VSP, LUKE for IRP, and LUKE with naïve score decoding (§6.3) or sequence sorting decoding (§6.4) for TRP. The following subsections describe our error analysis in the settings of independent prediction for each task and integrated pipeline prediction for the three tasks.

6.1 Visit Status Prediction

As stated in §5.2, the LUKE baselines often misclassified mentions with the `UnkOrNotVisit` label and entities with the `Other` label. Our analysis reveals two error tendencies. First, LUKE sometimes failed to distinguish factual statements from descriptions of traveler's visitation. For example, the sentence 027-01 in Table 7 (document 23251) is a factual statement, not indicating the visitation, so the mention "Matsue Shinjiko Onsen Station" should have been assigned `UnkOrNotVisit`. Second, LUKE sometimes fails to correctly understand the meaning of some motion verbs, such as "skip" and "pass on." For example, the mentions "Matsue" and "Yonago" are clearly stated as locations that the writer did not visit, as shown in Table 7 (document 23251, sentence 025-02).

6.2 Inclusion Relation Prediction

The results shown in Table 5 (§5.3) have indicated that IRP is a challenging task. Our analysis reveals that LUKE learned the tendency that prefectures and cities often become parents of some entities, but LUKE also sometimes made incorrect predictions, such as a prefecture/city being the parent of another prefecture/city. For example, LUKE predicted "Nagoya" as the parent of "Ise," although both are cities (sentence 046 of document 20816

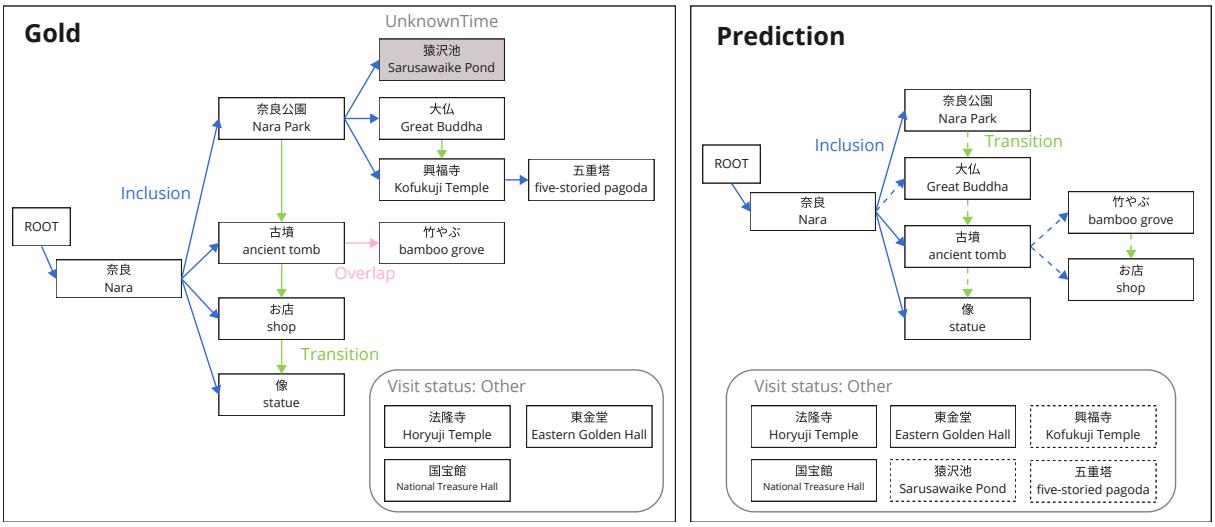


Figure 3: Illustrations of the gold visiting order graph and LUKE's prediction for an actual document (ID 00019). The nodes with dashed frames and the edges with dashed arrows represent prediction errors.

SentID	Original Japanese Text	English Translation
005	G:Vis P:Vis ~	The Great Buddha G:Vis P:Vis was really huge.
009	G:UN P:See G:Vis P:See G:Vis P:See	It's a photo of the five-storied pagoda G:Vis P:See at Kofukuji Temple G:Vis P:See visible from Sarusawaike Pond G:Vis P:See .
018	G:Vis P:Vis	When entering the ancient tomb G:Vis P:Vis , you can perceive the size more clearly.
019	G:Vis P:Vis	It is quietly situated in a bamboo grove G:Vis P:Vis .
021	G:Vis P:Vis	This is a shop G:Vis P:Vis famous for its persimmon-leaf sushi.

Table 8: Actual sentences in document 00019 with gold visit status ("G") and that prediced by LUKE ("P").

in Table 7). This suggests that the model lacks commonsensical geographic knowledge.

6.3 Transition Relation Prediction

The results shown in Table 6 (§5.4) have indicated difficulty in predicting reverse-order entity pairs. An example reverse pair is in sentences 024 and 025 (document 20851) in Table 7. While "Daiouji Temple" precedes "the station," these sentences describe that the traveler moved from the station to the temple. Although LUKE tended to predict the correct order of reverse pairs when there were some clues, such as temporal expressions like "before" and "after," but the system made incorrect predictions for reverse pairs without salient clues, including the above example.

6.4 Pipeline Prediction

Figure 3 shows the gold visiting order graph and prediction from the baseline systems[17] connected in the pipeline for document 00019.

For VSP, LUKE made correct predictions for 10 out of 13 entities. The three errors were caused by the predictions for three mentions in sentence 009 in Table 8; LUKE incorrectly predicted the mention label `See`, and then the MLA rule determined the entity label `Other`. LUKE failed to grasp the nuanced meaning of the sentence, which describes a photo of the facilities ("five-storied pagoda" and "Kofukuji Temple") taken by the traveler and the nearby location ("Sarusawaike Pond").

For IRP, LUKE predicted correct parents for four out of seven entities with the predicted label `Visit` and incorrect parents for the remaining three entities. Two of the failed entities are written

[17]They correspond to the systems stated at the beginning of §6. We simply refer to each system as "LUKE" here.

with general noun mentions ("bamboo grove" in sentence 019 and "shop" in sentence 021), and the geographic relations regarding these locations are not explicitly described. Therefore, the parents of them ("Nara") should be determined based on the consistent context of the traveler's trip to Nara. For correct prediction for another failed entity regarding the mention "Great Buddha" in sentence 005, which refers to famous Birushana Buddha at Todaiji Temple, geographic knowledge that Todaiji Temple is located in Nara Park is also necessary.

For TRP, LUKE was able to identify no exact entity pairs with correct transition relation mainly because the incorrect inclusion hierarchy was given as an input. This result indicates that the more accurate prediction of inclusion relation is crucial for the accurate prediction of transition sequences. However, LUKE successfully predicted many indirect transition relations, for example, "Nara Park," "ancient tomb," and "stature" have been visited in this order as LUKE also predicted. This suggests more lenient evaluation metric would also be useful in distinguishing such reasonable predictions from severe errors, such as pairs arranged in the reverse order from the correct visiting order.

7 Related Work

7.1 Visit Status Prediction

Visit status prediction can be regarded as a task in the field of human movement analysis for text, which includes a major stream of *predicate-centric* analysis and another *location-centric* analysis.

Predicate-Centric Analysis Represented by SpaceBank, a line of studies have sought to develop computational models that can recognize, generate and reason about spatial information, including place names, topological relations, and human movement (Pustejovsky et al., 2012; Pustejovsky and Yocum, 2013; Pustejovsky et al., 2015). These studies basically focused on verbs as expressions of movement. Similarly, previous event analysis studies from temporal or factuality perspectives treated verbs or predicates as a trigger of each event and specified attribute information on verbs, as represented by TimeML (Pustejovsky et al., 2003) and FactBank (Saurí and Pustejovsky, 2009). In user-generated contents including travelogues, geographic movements are often expressed without predicates, for example, *scene transition* by changing paragraphs or posts. Thus, we focus on geo-entities and their mentions, in contrast to the above predicate-centric analysis studies.

Location-Centric Analysis Some previous studies focused on "mention-level" visit status of location mentions in SNS posts (Li and Sun, 2014; Matsuda et al., 2018) and clinical documents (Peterson et al., 2021), which can be regarded as location-centric analysis. Long documents like travelogues often contain multiple mentions referring to the same location (i.e., geo-entity), and such different mentions can have different visit status labels. This necessitates a step of aggregating various visit status labels of mentions into the final visit status of the entity, which indicates whether the traveler eventually visited the location or not. Therefore, we additionally address "entity-level" prediction.

7.2 Visiting Order Prediction

Compared to many existing studies on the extraction of geo-entity mentions or toponyms (Lieberman et al., 2010; Matsuda et al., 2017; Kamalloo and Rafiei, 2018; Wallgrün et al., 2018; Weissenbacher et al., 2019; Gritta et al., 2020; Higashiyama et al., 2023), a few studies extracted some types of geographic *trajectories* from text (Ishino et al., 2012; Wagner et al., 2023; Kori et al., 2006).

Ishino et al. (2012) proposed a task and a method to extract transportation information, i.e., the origin and destination with a transportation method, from each disaster-related tweet. Such information can be regarded as part of a movement trajectory. Wagner et al. (2023) proposed a task to extract an entire trajectory from each transcription of testimony videos. Each testimony segment, which typically corresponds to one minute of the original video, was assigned a predefined location category, such as "`cities in Austria`" and "`ghettos in Hungary`." That is, their trajectory is a transition sequence of a coarse-grained location mainly mentioned in each scene, not a detailed movement trajectory of specific locations. Kori et al. (2006) proposed a method to extract and summarize sequences of location mentions as visitors' representative trajectories in blogs. They determined the visiting order of locations based on the order that mentions occurred in blogs. In contrast to Kori et al. (2006), we aim to identify

the faithful visiting order, which aligns with the written intentions. The crucial difference between these three studies and ours is the trajectory representation; we adopt not *sequences* but *graphs* as appropriate representations that retain geographic hierarchy as well as the temporal order of locations.

8 Conclusion

This study has proposed a *visiting order graph* to represent non-linear relations of visited locations and constructed an annotated travelogue dataset for graph-structured trajectory extraction. The experiments using the dataset have indicated the accuracy of baseline systems with some directions for performance improvement. The error analysis of actual examples has revealed notable tendencies of the prediction errors. One future direction is to develop sophisticated systems by addressing the issues we have raised for improvement. Another direction is to develop an integrated system for both trajectory extraction and grounding, which extracts an trajectory from each input document and associates its locations with points/areas of a map.

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A Annotation Examples

1	Visit	! (Arrived safely at the summit of Mount Akadake!)
2	Plan	 (Heading to Hotaka Shrine.)
3	See	(Mount Iodake is becoming visible nearby.)
4	Past	3~4 (About 34 years ago at the Naniwa Gyoza Stadium)
5	Future	(Next time, I'd like to visit Matsue and Yonago.)
6	UnkNot	1 (The train bound for Itoigawa Station consists of one car.)

Table 9: Example mentions annotated with visit status labels. Plan, Past, Future, and UnkNot indicate PlanToVisit, Visit-Past, Visit-Future, and UnkOrNotVisit, respectively.

B Detailed Dataset Statistics

Set	Visit	Plan	See	Past	Fut	UN/O
Train	1,942	–	–	–	–	397
Dev	252	–	–	–	–	64
Test	575	–	–	–	–	124
Train	2,577	358	212	10	6	619
Dev	332	48	46	1	4	74
Test	748	121	59	10	4	160

Table 10: Numbers of visit status labels for entities (top) and mentions (bottom). Plan, Past, and Fut indicate PlanToVisit, Visit-Past, and Visit-Future, respectively. UN/O indicates UnkOrNotVisit for mention level and Other for entity level.

Set	Inc	Trans	Over	UnkTime	MV
Train	1,302	1,041	38	35	95
Dev	186	143	8	8	16
Test	375	322	5	10	32

Table 11: Detailed statistics for visiting order annotation. Inc (Inclusion), Trans (Transition), and Over (Overlap) indicate the numbers of entity pairs with each relation type. UnkTime (UnknownTime) indicates the number of entities with the label. MV indicates the number of entities with multiple visits.

C Model Training Settings

Parameter	VSP	VOP
Training epochs	10	10
Batch size	16	8
Gradient accumulation step	4	1
Learning rate	5e-6	5e-6

Table 12: Hyperparameters of LUKE models for mention-level visit status prediction (VSP) and visiting order prediction (VOP).

D Detailed Experimental Results

D.1 Visit Status Prediction

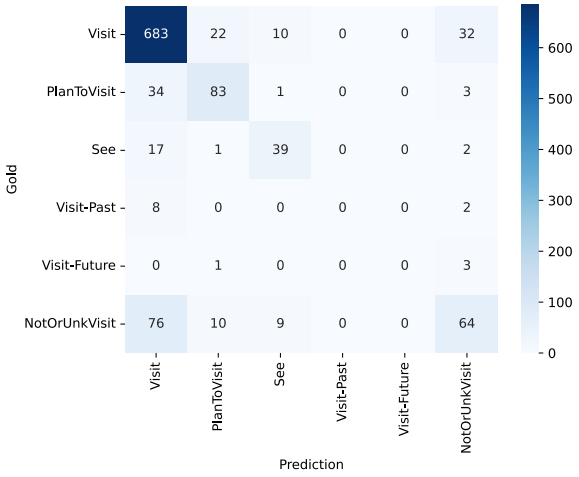


Figure 4: Confusion matrix of LUKE for mention-level visit status prediction.

Ablation Study To investigate the influence of surface text on learning and prediction of the baseline model for mention-level VSP, we evaluated two additional variants of the LUKE baseline trained with edited input text. That is, (1) "mention masking" model trained with input text where mention tokens are replaced by [MASK] tokens, and (2) "mention only" model trained with input text where context tokens are removed. Table 13 shows the performance of the model variants on the development set. Compared to the original baseline, the mention masking model remained slightly lower in accuracy (-0.024 points), and the mention only model, while even lower in accuracy (-0.145 points), was still able to predict correct labels to some extent. This suggests that the model mainly relied on context information and also used mention information together.

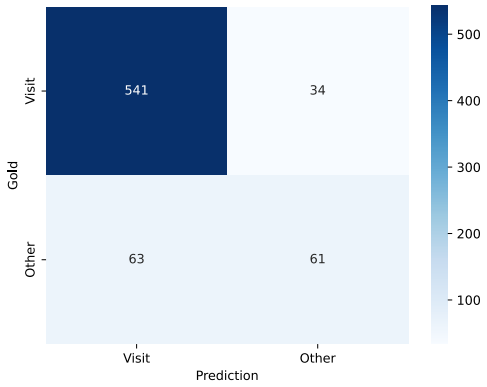


Figure 5: Confusion matrix of LUKE for entity-level visit status prediction.

Method	Acc.	Macro F1
LUKE	0.817	0.486
LUKE (mention masking)	0.793	0.457
LUKE (mention only)	0.672	0.205

Table 13: Performance of LUKE variants for mention-level visit status prediction on the development set.

D.2 Inclusion Relation Prediction

Depth	#Ent	F1		
		Random	Flat	LUKE
1	114	0.057	**1**	0.058
2	194	0.040	0	**0.432**
3	111	0.035	0	**0.438**
4	42	0.034	0	**0.305**
5	7	0.034	0	**0.743**

Table 14: System performance (F1 score) for inclusion relation prediction for each depth. Depth indicates the distance to `ROOT` of entity nodes based on the gold inclusion hierarchy.

D.3 Transition Relation Prediction

Size	#	F1			
		Rand.	Occ-V	LUKE-N	LUKE-S
2	34	0.498	**0.941**	0.782	0.919
3	36	0.332	**0.861**	0.744	0.845
4	21	0.245	0.667	0.685	**0.732**
5	24	0.196	0.750	0.754	**0.808**
6	35	0.165	**0.857**	0.765	0.847
7	12	0.138	0.750	0.435	**0.517**
8	35	0.126	**0.629**	0.549	0.566
9	32	0.108	**0.812**	0.718	0.800
$\geq$10	93	0.069	0.667	0.627	**0.681**

Table 15: System performance for transition relation prediction for each size of candidate entity sets. OccV represents OccOrder with the visit status strategy. LUKE-N and LUKE-S represent LUKE with naïve score-based decoding and sequence sorting decoding, respectively.